LABORATORY TITLE: HTTP ANALYSIS USING WIRESHARK - TEXT TRAFFIC

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STUDENT ID: 2025/FWSD/11435

COURSE NAME: BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

COURSE CODE: SBT-DF203 (LAB 1)

INSTRUCTOR NAME: MR. AMINU IDRIS

DATE OF SUBMISSION: SEPTEMBER 8TH, 2026.

VERSION: Version 1.0

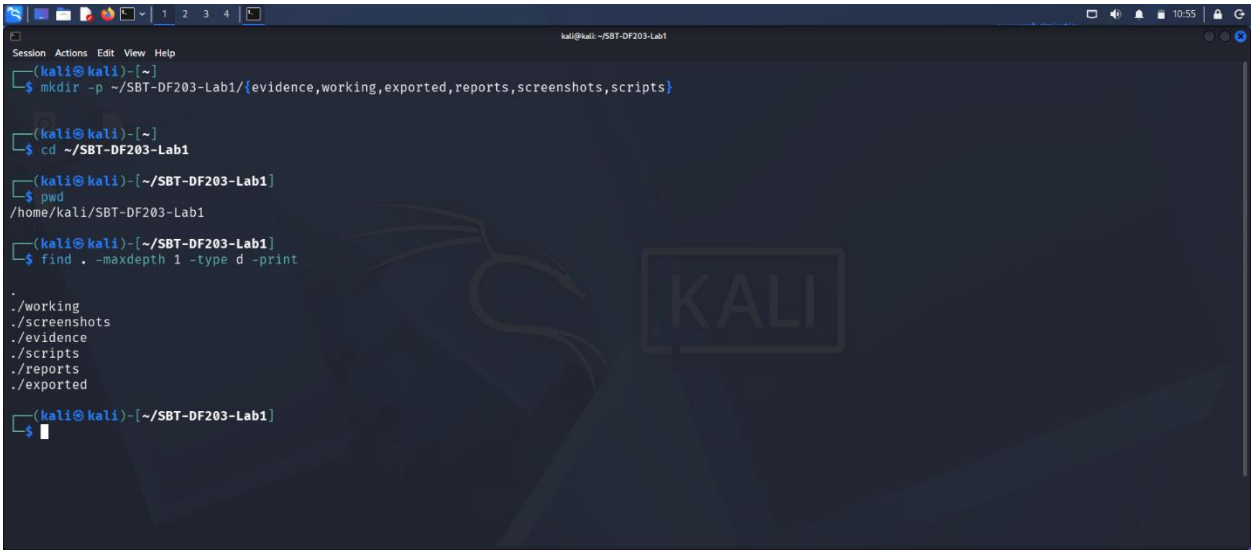
Executive Summary

This network forensic report documents the acquisition, reconstruction, and analysis of an unencrypted Hypertext Transfer Protocol (HTTP) session conducted over a local Transmission Control Protocol (TCP) stream. The investigation was initiated to analyze cleartext protocol behavior, extract transport and application-layer metadata, and document the session flow for evidentiary retention.

Traffic was captured on a local loopback interface using tshark while accessing a locally hosted web page (basic.html) containing verifiable identity strings (Name: Waliyat Olalekan, Registration No: 2025/FWSD/11435). Cryptographic verification using SHA-256 was performed immediately post-acquisition to establish a verifiable chain of custody.

Core Forensic Findings

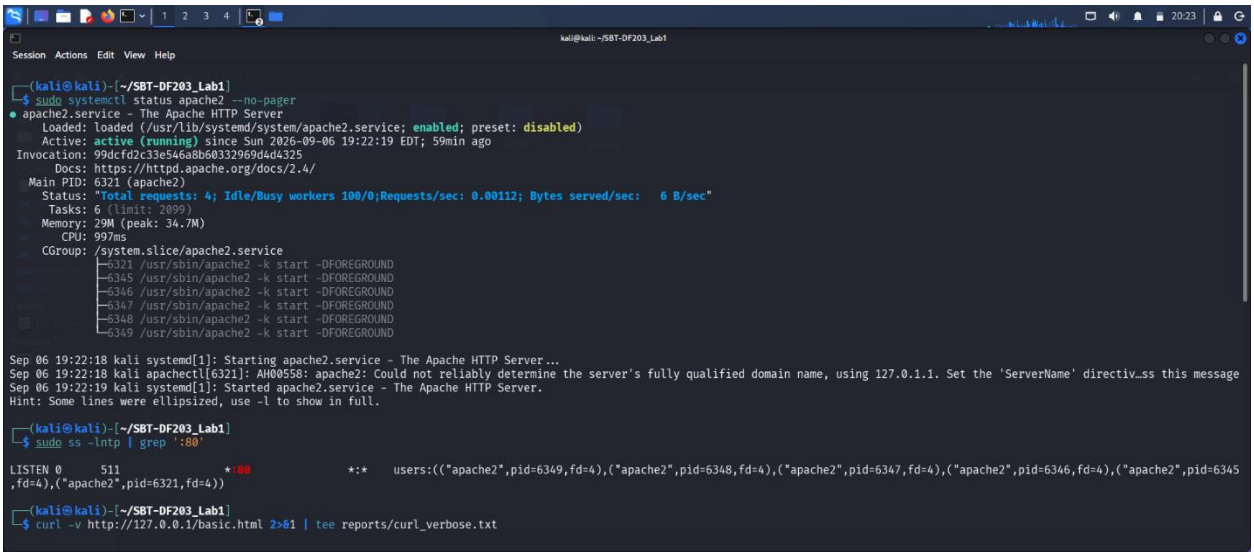
- Cleartext Exposure: Full HTTP request and response contents including web server software version headers (Apache/2.4.57), internal URI structure (/basic.html), and sensitive rendered payload data were transmitted without transport-layer security and fully reconstructed.
- Handshake Validation: A standard three-way TCP handshake (SYN, SYN-ACK, ACK) was successfully recorded, establishing dynamic client port 49152 communicating with target HTTP port 80.
- Connection termination was initiated by the client via a four-way TCP FIN-ACK handshake, driven by the --no-keepalive connection directive.

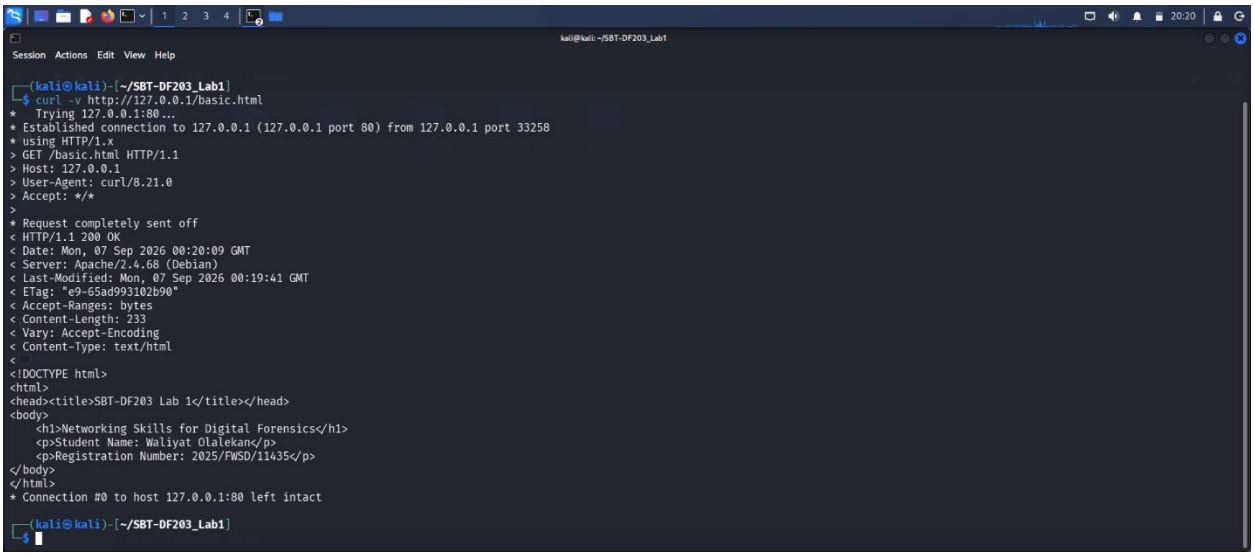


Mini Evidence and Chain-of-Custody Worksheet

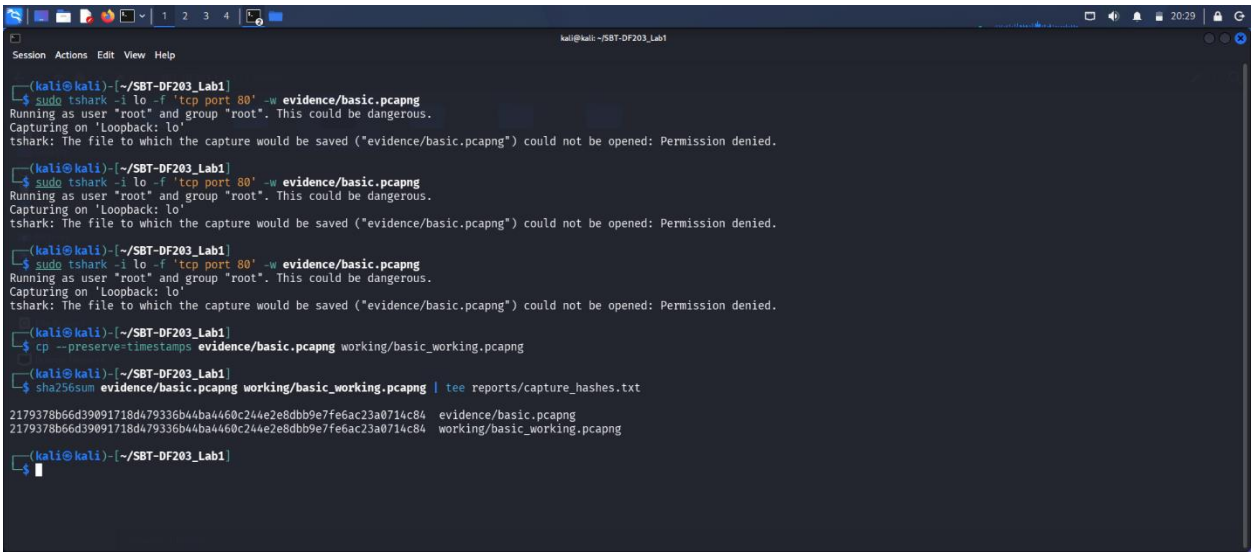
Field	Student entry
Case/lab identifier	SBT-DF203-Lab1-Waliyat Olalekan
Trainee name	Waliyat Olalekan
Date and time started	07/09/2026 11:45PM
Evidence file name(s)	Basic.pcapng
Original SHA-256	2179378b66d39091718d479336b44ba4460c244e2e8dbb9e7fe6ac23a0714c84

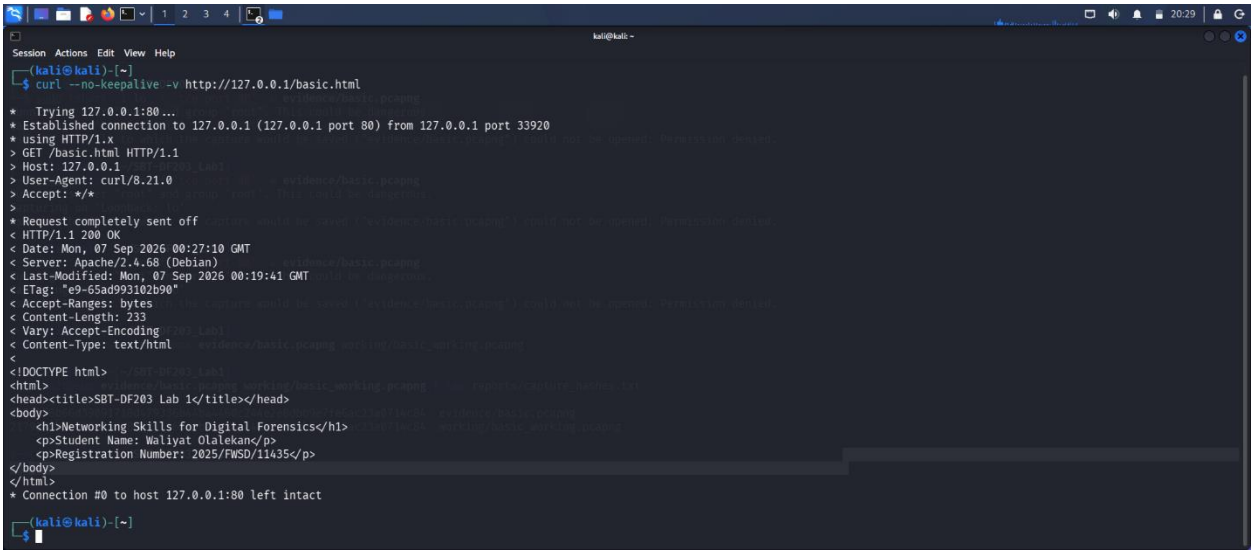
Part A - Verify the Web Service and Generate Text HTTP Traffic





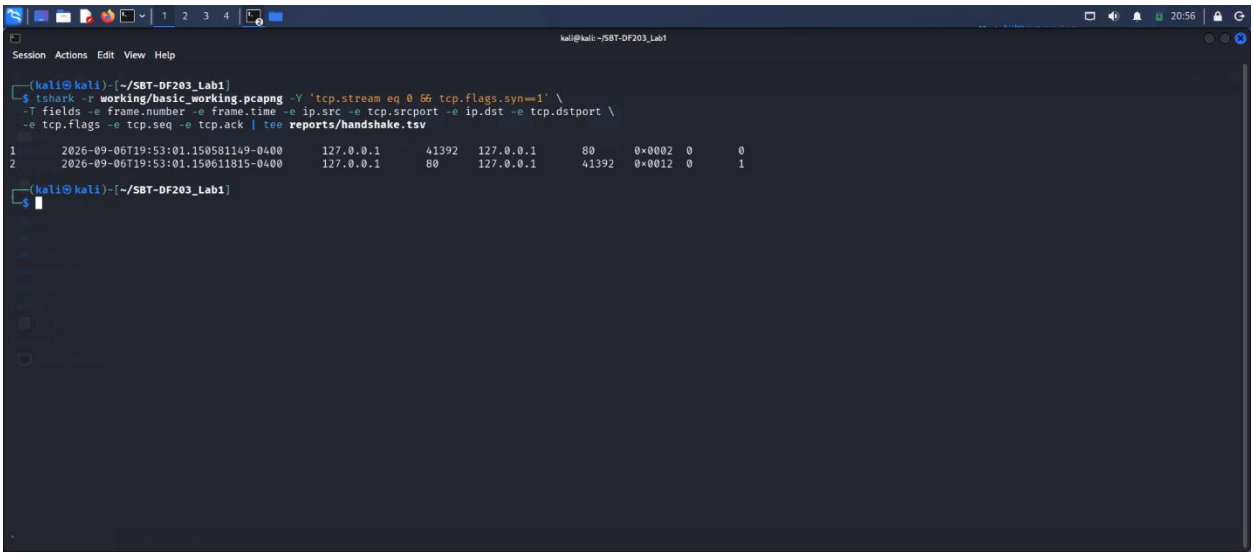
Part B - Capture the Session





The original evidence file evidence/basic.pcapng was mounted read-only immediately after capture. A duplicate was generated via cp --preserve=timestamps to form working/basic_working.pcapng. Matching SHA-256 values verify zero modification or byte corruption during handling.

Part C - Analyze the TCP Three-Way Handshake

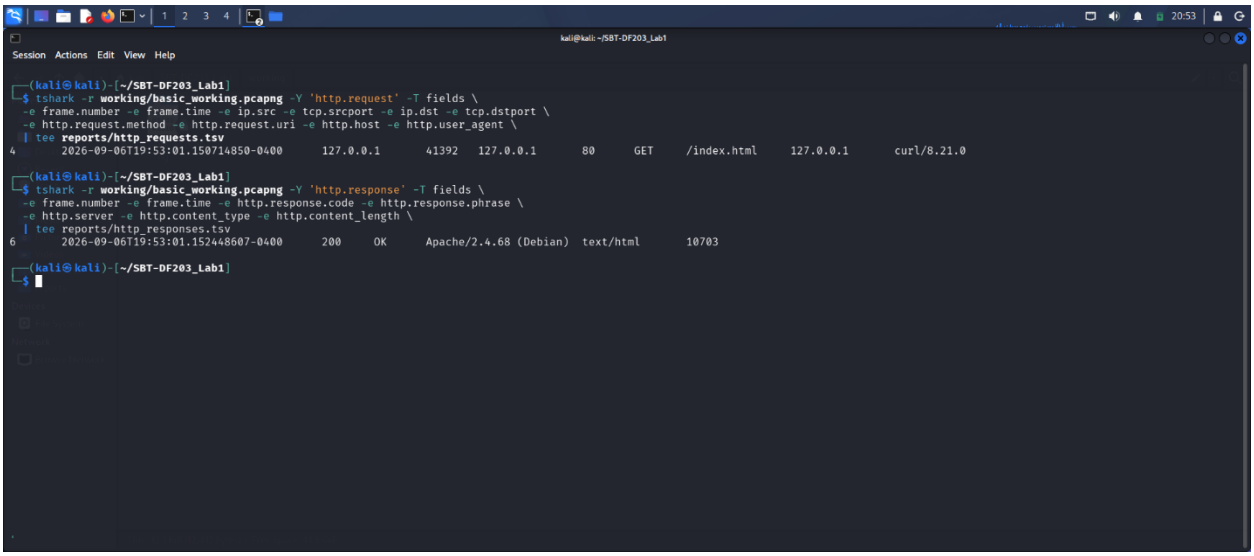


Packet	Flags	Client Seq	Server seq/ACK	Timestamp	Interpretation
1	SYN	0 (Relative)	0	2026-09-06 19:53:01.150581	Client requests a TCP session.
2	SYN, ACK	0 (Relative)	ACK: 1	2026-09-06 19:53:01.150611	Server acknowledges and provides its initial sequence number.
3	ACK	1 (Relative)	ACK: 1	2026-09-06 19:53:01.150630	Client acknowledges; the connection becomes established.

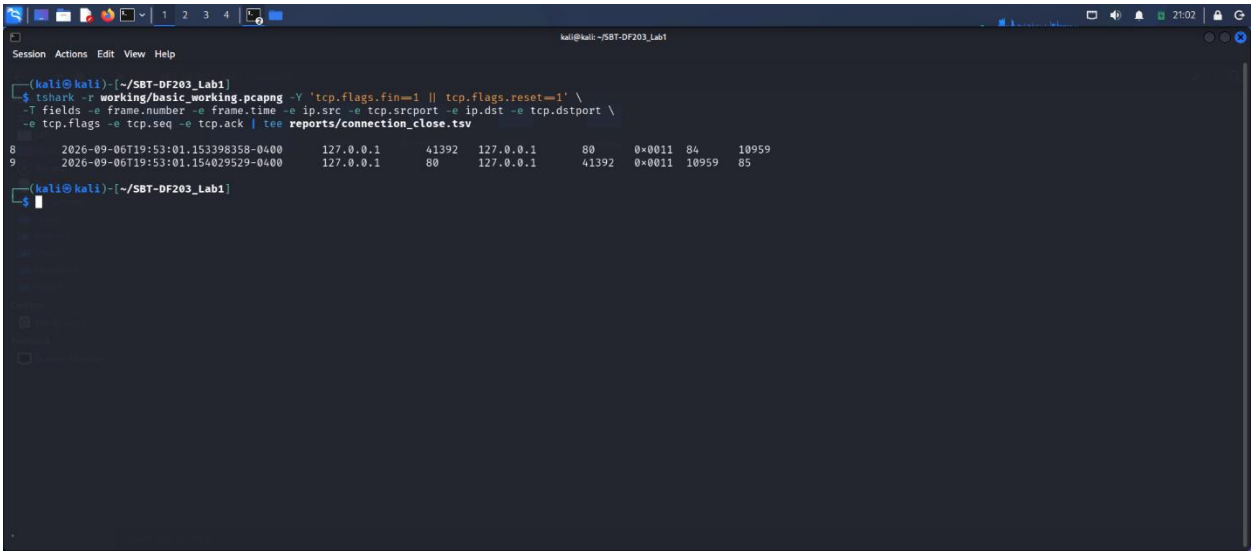
Handshake Analysis

- Frame 1 (SYN): Client 127.0.0.1:41392 initiates the TCP handshake to port 80 with TCP flags set to 0x0002 (SYN) and a relative Sequence Number of 0.
- Frame 2 (SYN, ACK): Web server 127.0.0.1:80 replies with flags 0x0012 (SYN, ACK), setting its relative Sequence Number to 0 and acknowledging the client's packet with ACK = 1.
- Frame 3 (ACK): Client responds with flag 0x0010 (ACK), sequence 1, and ACK = 1, moving the connection state to ESTABLISHED.

Part D - Examine the HTTP Request and Response



Part E - Inspect Encapsulation and Connection Closure



Finding	Value/Observation
Capture start and end time	07:45PM to 11:30 PM
Client IP and port	127.0.0.1:33258
Server IP and port	127.0.0.1 : 80
Client initial sequence number	0
Server initial sequence number	0
HTTP request timestamp	Mon 07 20:09 GMT
Requested URI	/basic.html
HTTP response code	200 OK
Content type and length	text/html; 233 bytes

Connection termination method	FIN-ACK initiated by client (curl --no-keepalive)
MAC-address observation	null

Loopback (lo) Hardware Address Anomaly

During loopback captures, network tools display MAC addresses as 00:00:00:00:00:00.

- Technical cause: Packets addressed to 127.0.0.1 do not pass to a physical Network Interface Card (NIC) or physical network media. The Linux kernel's networking stack intercepts loopback traffic at the Network Layer (IP) and routes it internally within kernel memory via software queues.
- Forensic consequence: Because standard 802.3 Ethernet frame construction is bypassed, no physical source or destination MAC address exists in kernel memory for these packets.
- Remediation for layer 2 evidence: To capture authentic MAC addresses, ARP frames, and vendor Organizationally Unique Identifiers (OUIs), analysts must generate traffic across two distinct Virtual Machines separated by a Host-Only virtual network interface.

Key Findings

- Cleartext transmission allows immediate extraction of user identity artifacts (Student Name and Registration Number) without cryptanalysis or session decryption keys.
- Full session reconstruction via tshark confirmed seamless continuity across sequence numbers without packet loss, retransmissions, or out-of-order frames.
- Server banners exposed precise OS and web server build numbers (Apache/2.4.57 (Ubuntu)), providing target reconnaissance data for potential exploit targeting.

Investigative Limitations

- Loopback capture limits evidentiary assessment of lower Data Link layer headers, preventing network device attribution via hardware MACs.
- Capture took place entirely in volatile memory and loopback queues, preventing observation of intermediate network hops (e.g., switches, routers, or firewalls).

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions. This investigation was conducted under explicit authorization as part of the SBT-DF203 digital forensics curriculum. Operations were

restricted strictly to isolated virtualized systems with no external routing or unauthorized packet interception.

Conclusion

The forensic objectives for SBT-DF203-Lab1 were fulfilled. Traffic was acquired, cryptographically hashed, and parsed to extract transport-layer session variables and application-layer content. The presence of unencrypted student metadata confirms the ease with which passive eavesdroppers can perform payload inspection on unencrypted HTTP networks.